STAT-251: Probability and Statistics — Worked Solutions

Jahangirnagar University, Part II Semester II, 2021

Q1 — Mode and Karl Pearson's Coefficient of Kurtosis

Concepts

Mode (grouped data). The mode is the value that occurs most often. For grouped/continuous data we identify the **modal class** (highest frequency) and interpolate within it:

$$\text{Mode} = L + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h$$

- L = lower boundary of the modal class
- f_1 = frequency of the modal class
- f_0 = frequency of the class *before* it
- f_2 = frequency of the class *after* it
- h = class width

Kurtosis. Kurtosis measures how *peaked* a distribution is relative to a normal distribution (i.e., how heavy its tails are). Karl Pearson's coefficient uses the 4th central moment:

$$\beta_2 = \frac{\mu_4}{\mu_2^2}, \qquad \gamma_2 = \beta_2 - 3 \text{ (excess kurtosis)}$$

where μ_2 and μ_4 are the 2nd and 4th moments about the mean. A normal distribution has $\beta_2 = 3$ ($\gamma_2 = 0$):

Result	Shape	Name
$\beta_2 > 3$ ($\gamma_2 > 0$)	Peaked, heavy tails	**Leptokurtic**
$\beta_2 = 3$ ($\gamma_2 = 0$)	Normal	**Mesokurtic**
$\beta_2 < 3$ ($\gamma_2 < 0$)	Flat, light tails	**Platykurtic**

For grouped data we compute moments efficiently with the **step-deviation method**: let $u = (x - A)/h$ for a convenient class midpoint A, then convert moments back afterward. Because β_2 is a *ratio* of μ_4 to μ_2^2, and both scale by h^4, the class width cancels — you can compute β_2 directly from the u-values.

Data (with midpoints and step deviations, $A = 2.225, h = 0.89$)

Class	Frequency f	Midpoint x	$u = (x - A)/h$
0.00–0.89	63	0.445	-2
0.89–1.78	128	1.335	-1
1.78–2.67	209	2.225	0
2.67–3.56	106	3.115	1
3.56–4.45	76	4.005	2
4.45–5.35	41	4.90	3
5.35–6.24	19	5.795	4
6.24 and above	12	6.685	5

$N = \sum f = 654$

(i) Mode

Modal class = **1.78–2.67** (highest frequency, $f_1 = 209$), with $f_0 = 128, f_2 = 106, L = 1.78$, $h = 0.89$:

$$\text{Mode} = 1.78 + \frac{209 - 128}{2(209) - 128 - 106} \times 0.89 = 1.78 + \frac{81}{184} \times 0.89 \approx \boxed{2.17 \text{ knots}}$$

(ii) Karl Pearson's coefficient of kurtosis

Computing $\sum fu, \sum fu^2, \sum fu^3, \sum fu^4$ from the table gives:

$$\sum fu = 263, \quad \sum fu^2 = 1763, \quad \sum fu^3 = 3905, \quad \sum fu^4 = 18143$$

Raw moments: $\bar{u} = 263/654 = 0.4021$

$$\mu'_2 = 1763/654 = 2.6957, \quad \mu'_3 = 3905/654 = 5.9694, \quad \mu'_4 = 18143/654 = 27.7416$$

Converting to central moments:

$$\mu_2 = \mu'_2 - \bar{u}^2 = 2.5340$$

$$\mu_3 = \mu'_3 - 3\bar{u}\mu'_2 + 2\bar{u}^3 = 2.8473$$

$$\mu_4 = \mu'_4 - 4\bar{u}\mu'_3 + 6\bar{u}^2\mu'_2 - 3\bar{u}^4 = 20.6766$$

(As a check: $\bar{x} = A + h\bar{u} \approx 2.58$ knots, $\sigma \approx h\sqrt{\mu_2} \approx 1.42$ knots.)

$$\beta_2 = \frac{\mu_4}{\mu_2^2} = \frac{20.6766}{(2.5340)^2} = \frac{20.6766}{6.4211} \approx \boxed{3.22}$$

$$\gamma_2 = \beta_2 - 3 \approx 0.22$$

(iii) Comment

Since $\beta_2 \approx 3.22 > 3$ (equivalently $\gamma_2 > 0$), the wind-speed distribution is **leptokurtic**: it is more sharply peaked around its mean and has heavier tails than a normal distribution with the same variance. This means wind speeds cluster tightly around the modal value (≈2.17 knots) more than a normal curve would predict, but extreme readings (both very calm and very high wind) are also somewhat more likely than under normality.

Q2 — Box-and-Whisker Plot, Skewness, and Outliers

Concepts

Five-number summary & the box plot. A box plot displays: minimum (non-outlier), Q_1, median, Q_3, maximum (non-outlier), plus outliers as separate points. With n values sorted:

- **Median** = middle value (average of two middle values if n is even)
- Q_1 = median of the lower half
- Q_3 = median of the upper half
- $\text{IQR} = Q_3 - Q_1$ (the box width — spread of the middle 50%)

Outlier fences (Tukey's rule):

Type	Lower fence	Upper fence
Mild outlier	$Q_1 - 1.5\,\text{IQR}$	$Q_3 + 1.5\,\text{IQR}$
Extreme outlier	$Q_1 - 3\,\text{IQR}$	$Q_3 + 3\,\text{IQR}$

A point beyond the *mild* fence but within the *extreme* fence is a mild outlier; beyond the extreme fence, it's an extreme outlier. Whiskers extend only to the most extreme **non-outlier** data points.

Skewness from a box plot. Compare the two halves of the box, and the whisker lengths: if the box/whisker/outliers stretch further above the median than below, the data is **right (positively) skewed**, and vice versa for left skew.

Sorted data (n = 32)

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1,1,3,4,5,8,17,23,24,24,25,25,25,26,26,27,28,28,28,29,29,29,29,30,35,37,40,45,4
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(i) Box-and-whisker plot

- Median = average of 16th & 17th values = (27+28)/2 = **27.5**
- Q_1 = median of lower 16 values = (23+24)/2 = **23.5**
- Q_3 = median of upper 16 values = (30+35)/2 = **32.5**
- $\text{IQR} = 32.5 - 23.5 = 9$

Fences:

- Mild: lower $= 23.5 - 13.5 = 10$, upper $= 32.5 + 13.5 = 46$
- Extreme: lower $= 23.5 - 27 = -3.5$, upper $= 32.5 + 27 = 59.5$

Whiskers run to the most extreme values **inside** the mild fences: lower whisker → **17**, upper whisker → **45**. *(See the box plot rendered above — box from 23.5 to 32.5, median line at 27.5, whisker from 17 to 45, low-side dots for the mild low outliers, and a diamond marker for the extreme outlier at 60.)*

(ii) Skewness

- Median − Q_1 = 27.5 − 23.5 = 4
- Q_3 − median = 32.5 − 27.5 = 5

The upper half of the box is wider, the upper whisker (32.5→45, length 12.5) is much longer than the lower whisker (17→23.5, length 6.5), and **all** the outliers sit on the high side. All three signals agree: the data is **right-skewed (positively skewed)** — most students waited a "typical" short time, but a few waited unusually long.

(iii) Mild and extreme outliers

- **Below the mild fence (< 10):** 1, 1, 3, 4, 5, 8 — all are above the extreme fence (−3.5), so these 6 values are **mild low outliers**.
- **Above the mild fence (> 46):** 53, 53, 60.
  - 53 and 53 are below the extreme fence (59.5) → **mild high outliers** (2 values).
  - 60 exceeds the extreme fence (59.5) → **extreme high outlier** (1 value).

Summary: Mild outliers = {1, 1, 3, 4, 5, 8, 53, 53} (8 values). Extreme outlier = {60} (1 value).

(iv) Portion of outliers in each category

$$\text{Mild outliers: } \frac{8}{32} = 0.25 = 25\%$$

$$\text{Extreme outliers: } \frac{1}{32} = 0.03125 \approx 3.1\%$$

$$\text{Total outliers: } \frac{9}{32} \approx 28.1\%$$

Q3 — Simple Linear Regression and Correlation

Concepts

Least-squares regression line of Y on X: $\hat{Y} = a + bX$, chosen to minimize $\sum(Y - \hat{Y})^2$:

$$b = \frac{n\sum XY - \sum X \sum Y}{n\sum X^2 - (\sum X)^2}, \qquad a = \bar{Y} - b\bar{X}$$

Pearson correlation coefficient r measures the strength/direction of the *linear* relationship ($-1 \le r \le 1$):

$$r = \frac{n\sum XY - \sum X \sum Y}{\sqrt{[n\sum X^2 - (\sum X)^2]\,[n\sum Y^2 - (\sum Y)^2]}}$$

Coefficient of determination, r^2, is the proportion of the variance in Y that is explained by the linear regression on X (the rest is unexplained/residual variance).

Data summary (n = 15)

$$\sum X = 798,\ \sum Y = 643,\ \sum XY = 51232,\ \sum X^2 = 63040,\ \sum Y^2 = 41999$$

$$\bar{X} = 53.2, \qquad \bar{Y} = 42.87$$

(a) Regression line to estimate runoff volume

$$b = \frac{15(51232) - (798)(643)}{15(63040) - 798^2} = \frac{768480 - 513114}{945600 - 636804} = \frac{255366}{308796} \approx 0.827$$

$$a = 42.87 - 0.827(53.2) \approx -1.13$$

$$\boxed{\hat{Y} = -1.13 + 0.827\,X}$$

Each additional m³ of rainfall is associated with about 0.83 m³ more runoff.

(b) Estimated runoff when rainfall $X = 50$

$$\hat{Y} = -1.13 + 0.827(50) = -1.13 + 41.35 \approx \boxed{40.2\ \text{m}^3}$$

(c) Correlation coefficient and coefficient of determination

$$n\sum X^2 - (\sum X)^2 = 308796, \qquad n\sum Y^2 - (\sum Y)^2 = 15(41999) - 643^2 = 629985 - 413449 =$$

$$r = \frac{255366}{\sqrt{308796 \times 216536}} = \frac{255366}{\sqrt{66{,}865{,}450{,}656}} \approx \frac{255366}{258584} \approx \boxed{0.988}$$

$$r^2 \approx \boxed{0.975}$$

Comment

$r \approx 0.99$ indicates a **very strong, positive linear relationship** between rainfall and runoff volume — as rainfall increases, runoff increases almost proportionally. $r^2 \approx 0.975$ means about **97.5% of the variation in runoff volume is explained** by the linear regression on rainfall volume, leaving only ~2.5% due to other factors or random variation. This is a strong justification for using the linear model to estimate runoff from rainfall, as done in part (b).

Quick reference — key formulas used

Concept	Formula
Mode (grouped)	$L + \dfrac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h$
Kurtosis coefficient	$\beta_2 = \mu_4 / \mu_2^2$; excess $= \beta_2 - 3$
Quartiles	Q_1, Q_3 = medians of lower/upper halves
IQR	$Q_3 - Q_1$
Mild outlier fence	$Q_1 - 1.5\,\text{IQR}$ to $Q_3 + 1.5\,\text{IQR}$
Extreme outlier fence	$Q_1 - 3\,\text{IQR}$ to $Q_3 + 3\,\text{IQR}$
Regression slope	$b = \dfrac{n\sum XY - \sum X \sum Y}{n\sum X^2 - (\sum X)^2}$
Correlation coefficient	$r = \dfrac{n\sum XY - \sum X \sum Y}{\sqrt{[n\sum X^2 - (\sum X)^2][n\sum Y^2 - (\sum Y)^2]}}$
Coefficient of determination	r^2